

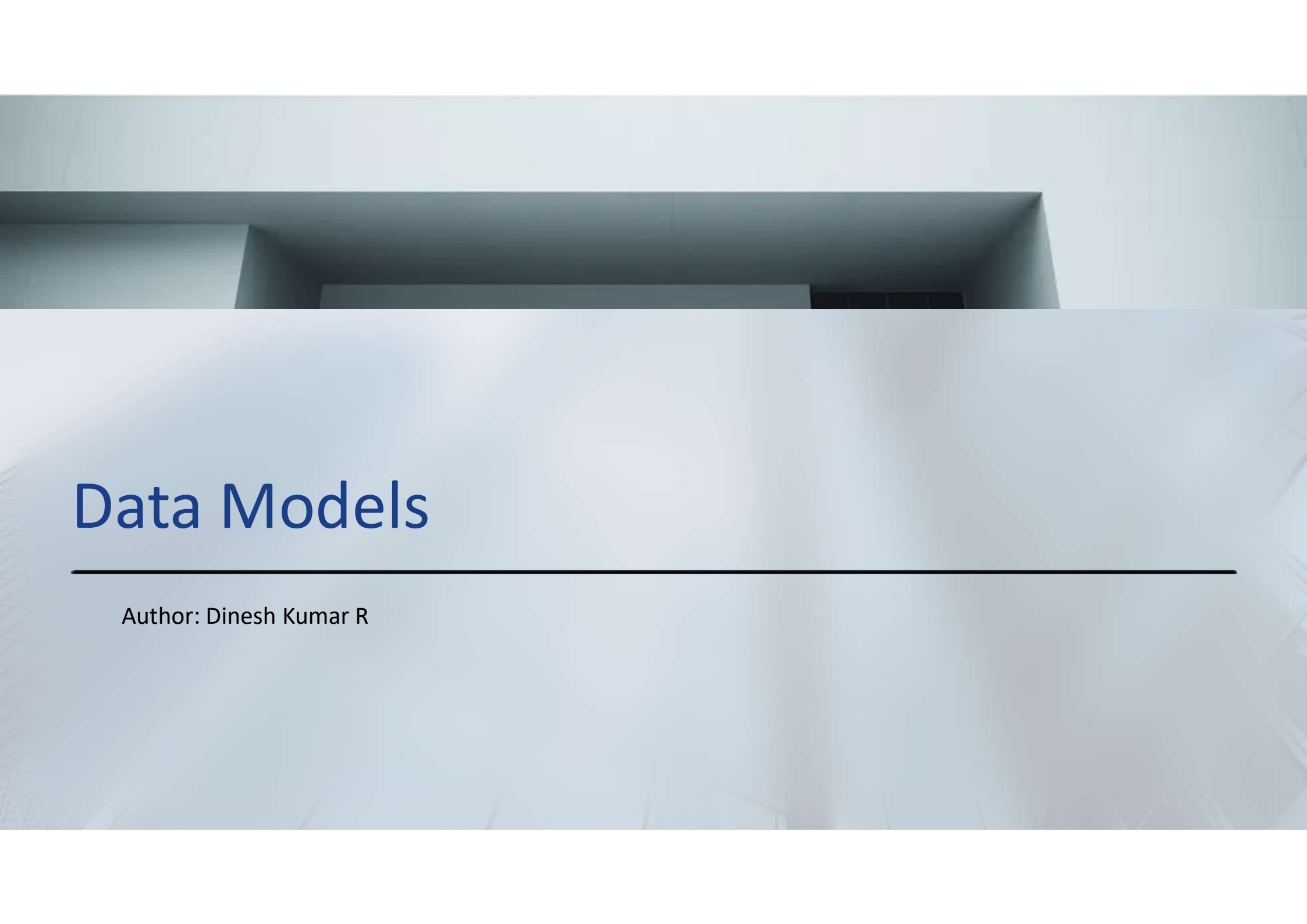


SSS COLLEGE OF ARTS, SCIENCE & MANAGEMENT
DEPARTMENT OF COMPUTER SCIENCE
ORACLE DATABASE FOUNDATION



REGISTER NO : 32723U18015

STUDENT NAME : R Dinesh Kumar



Data Models

Author: Dinesh Kumar R

CONTENTS

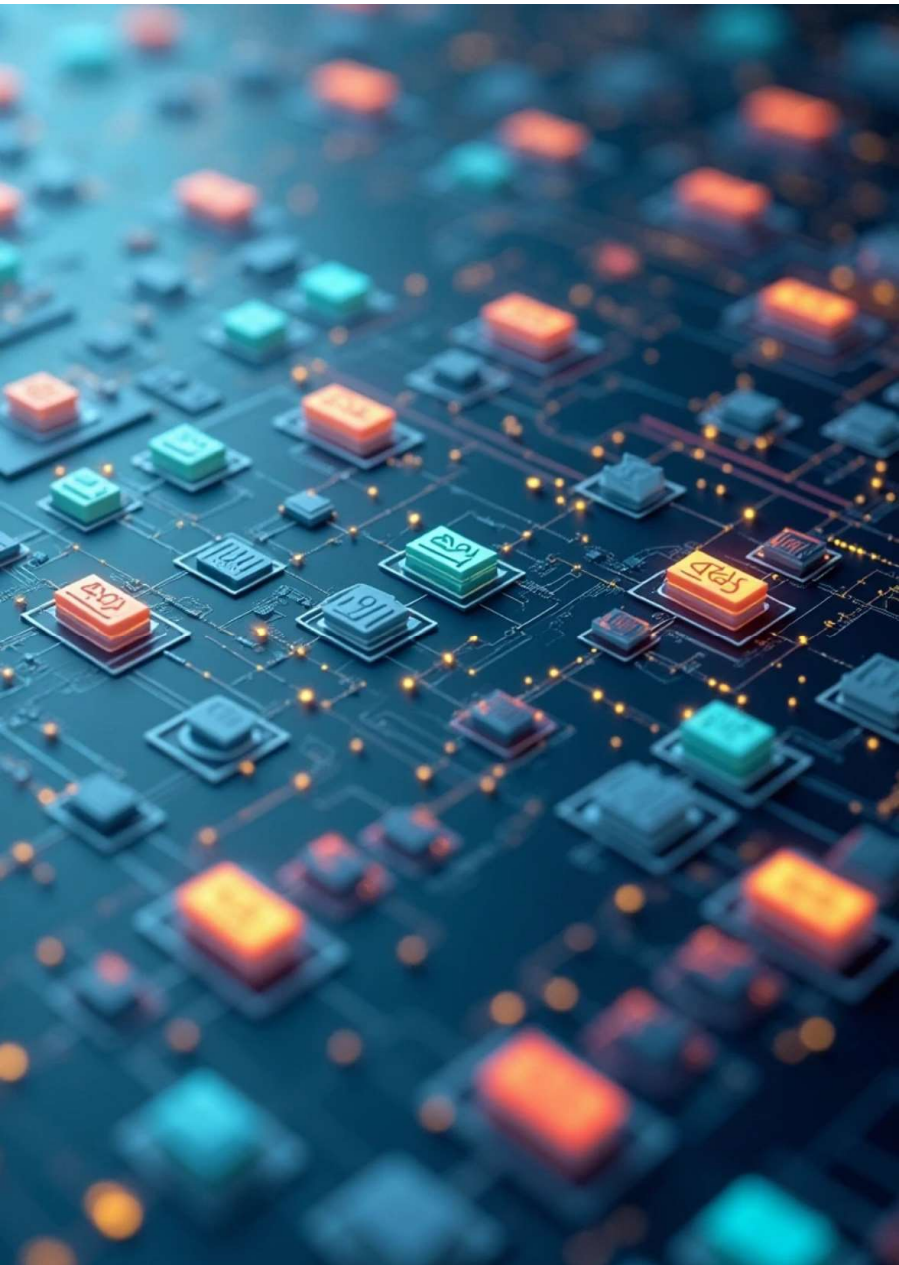
1. Overview of Data Models

2. Hierarchical Model

3. Network Model

4. Relational Model

5. Object-Oriented Model



1. Overview of Data Models

Data models describe the logical structure of a database, introducing abstraction in Database Management Systems (DBMS). They define how data is interconnected, processed, and stored within the system. Historically, three major models have been utilized: the hierarchical model, the network model, and the relational model.

2. Hierarchical Model

Structure

In a hierarchical model, data is organized into a tree-like structure where each record has a single parent. A sort field maintains sibling records in a specific order.

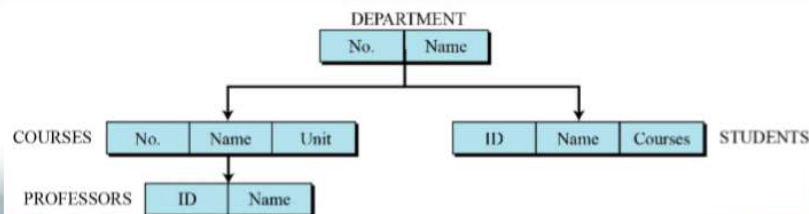
Relationships

This structure allows for one-to-many relationships, efficiently describing various real-world relationships like recipes and ordered information.

Access

Records are accessed through navigation using pointers and sequential access, which can lead to inefficiencies when full paths are not specified.

In the hierarchical model, data is arranged in an inverted tree. Each entity has one parent but can have multiple children, with a single root entity at the top.



3. Network Model

1

Expansion

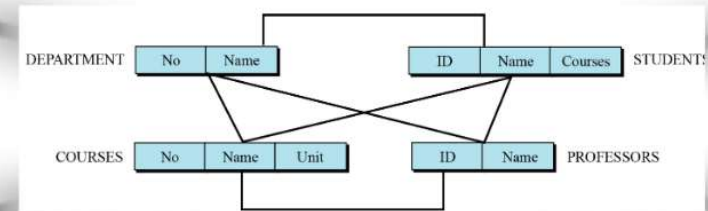
The network model allows many-to-many relationships within a tree-like like structure that supports multiple parents.

2

Access Methods

Records can be accessed sequentially or by navigating through circular linked lists.

In this model, entities are arranged in a graph, enabling access through multiple paths.



Relational Model

The **Relational Model** in **DBMS (Database Management System)** is one of the most widely used data models for organizing and managing data. It was proposed by **E.F. Codd** in **1970** at IBM. The model represents data in **tables (relations)**, making it intuitive and easy to use.

Popularity

The relational model is the most widely used data model in model in DBMS, based on first-order predicate logic where data where data is represented in tables (relations).

Key Highlights

- Data is organized in tables called relations.
- Relations can be normalized.
- Normalized relations ensure values are atomic.
- Each row in a relation contains unique values.
- Each column in a relation contains values from a single single domain.

Relational Model

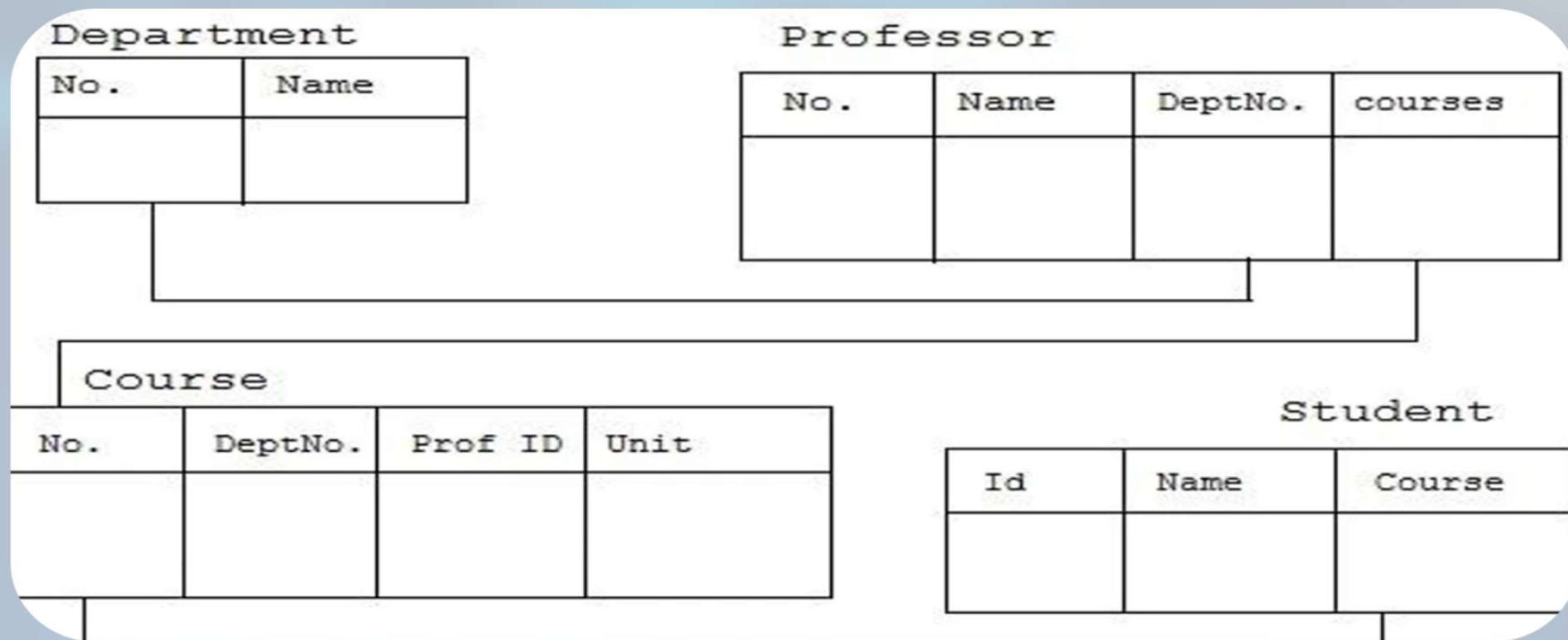


Fig 4.1

insert into RELATION-NAME
values (... , ... , ...)

COURSES

No	Course-Name	Unit
CIS15	Intro to C	5
CIS17	Intro to Java	5
CIS19	UNIX	4
CIS51	Networking	5

SQL Command

insert into COURSES
values ("CIS52", "TCP/IP" , 6)

COURSES

No	Course-Name	Unit
CIS15	Intro to C	5
CIS17	Intro to Java	5
CIS19	UNIX	4
CIS51	Networking	5
CIS52	TCP/IP	6

Fig 4.2

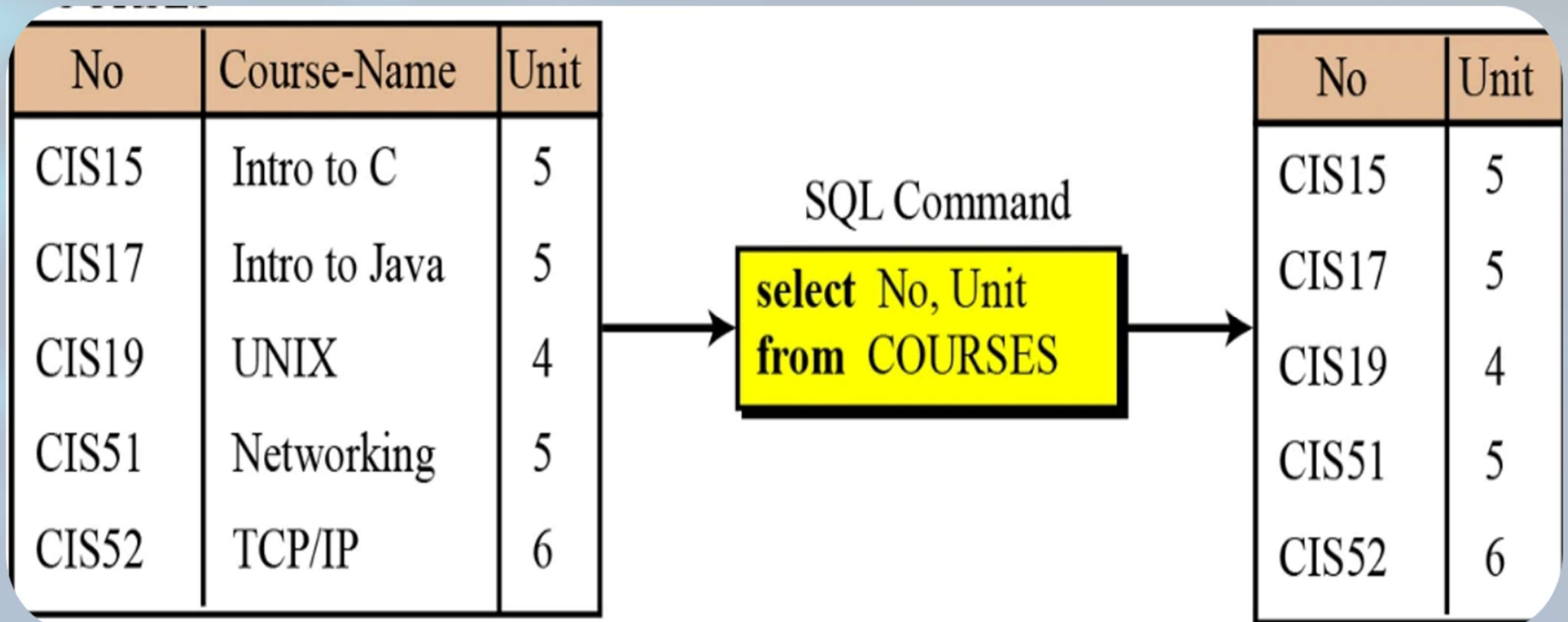


Fig 4.3

5. Object-Oriented Model

1

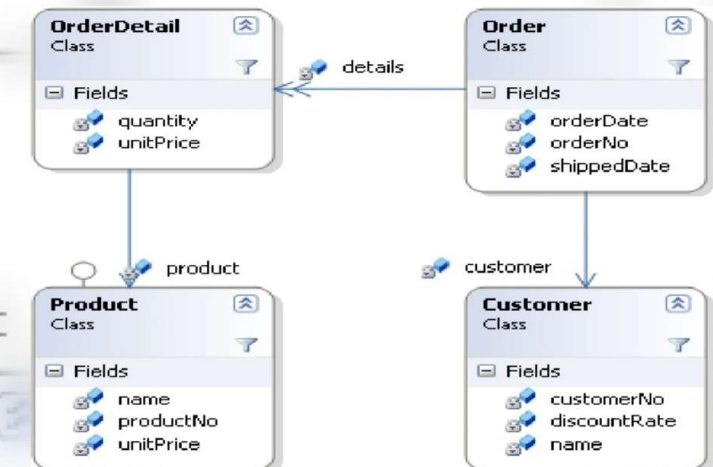
Introduction

In the 1990s, the object-oriented programming paradigm was integrated into database technology, leading to the creation of object databases.

2

Features

Object databases introduce core concepts of object programming, such as such as encapsulation and polymorphism.



Conclusion

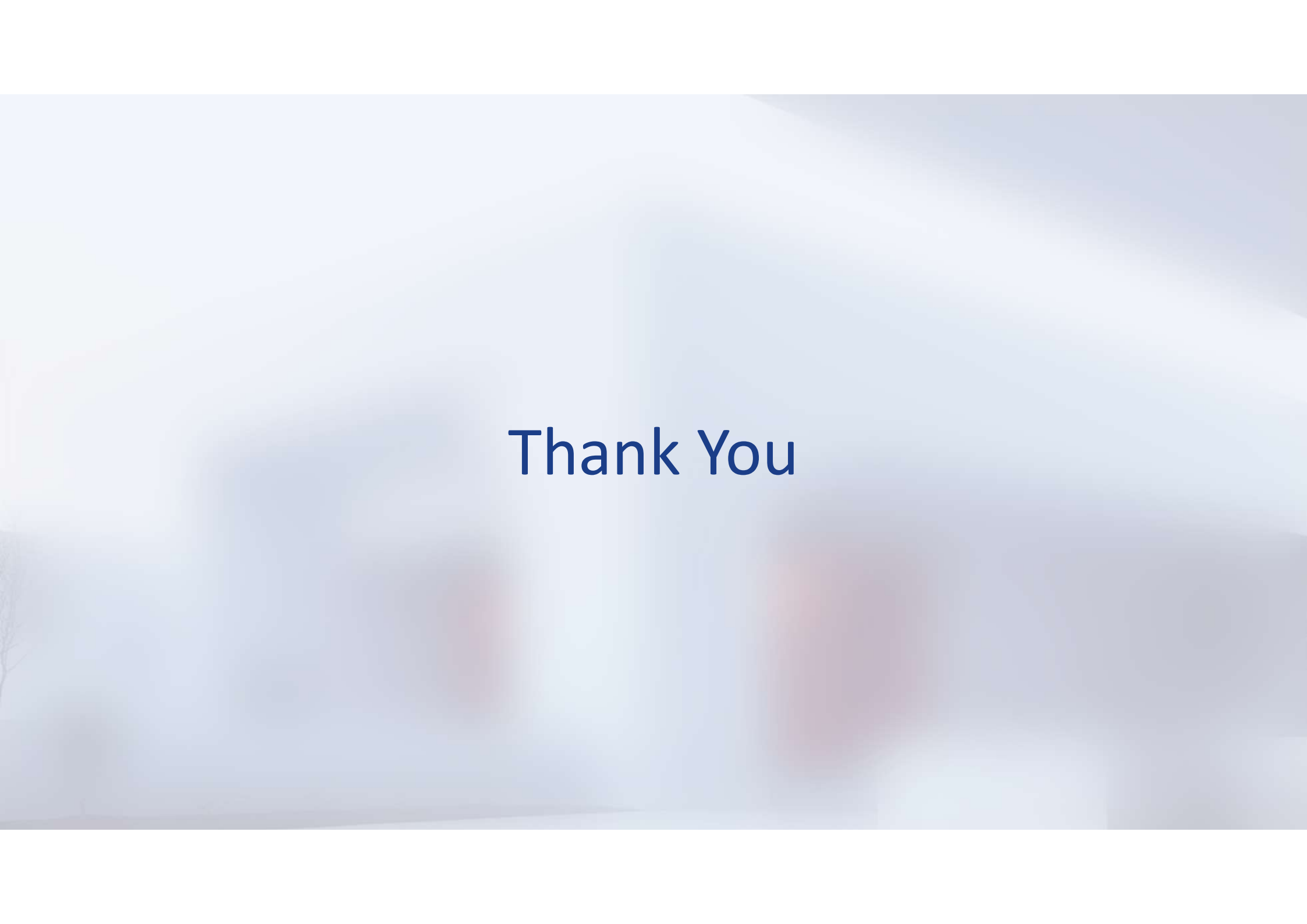
Data models are the foundation of any Database Management System.

They provide a structured framework for data organization, storage, and retrieval. Choosing the right data model (Hierarchical, Network, Relational, or Object-Oriented) is crucial for effective database design and performance.

Understanding data models helps in:

Ensuring data integrity
Supporting efficient queries
Enabling scalable and maintainable systems
As data continues to grow, modern DBMS are evolving with hybrid and NoSQL models to meet diverse application needs.

A well-chosen data model is key to unlocking the full potential of data in any organization.



Thank You